DE LA SALLE UNIVERSITY-DASMARINAS
COLLEGE OF INFORMATION AND COMPUTER STUDIES
INFORMATION TECHNOLOGY DEPARTMENT

COURSE SYLLABUS

COURSE CODE	**:**	**ITCP412**
COURSE TITLE	**:**	**Capstone Project 2**
COURSE TYPE	**:**	Lecture / Research
COURSE CREDIT	**:**	3 Units
LEARNING MODALITY	**:**	Traditional Blended Learning Model
PRE-REQUISITES	**:**	ITCP322 – Capstone Project 1
CO-REQUISITES	**:**	None
PROFESSOR	**:**	Paulino H. Gatpandan, DIT phgatpandan@dlsud.edu.ph
CONSULTATION HOURS	**:**	1000 – 1200 Tuesday (COS100C) 1000 – 1200 Wednesday (COS100C) 0900 – 1200 & 1300 – 1430 Friday (Online)

VISION

A leading Catholic university that inspires excellence and drives innovation toward a just, peaceful, and sustainable society.

MISSION

To champion the Human and Christian Education of lifelong learners who value history and culture through responsive and inclusive academic, research, and extension programs.

COURSE DESCRIPTION:

This course requires the fulfillment of the complete documentation, software, and oral defense before a set of panelists.

COURSE LEARNING OUTCOMES:

By the end of this course, students can:

CLO1. Understand the latest concepts in systems analysis and design, software engineering and information technology.

CLO2. Analyze original research data and present the data in a form appropriate for the

design.

CLO3. Collect original data following ethical protocols.
CLO4. Interpret the data collected, draw conclusions solidly rooted in the data, and convey those interpretations and implications in a formal written report.
CLO5. Design and implement integrated information systems and technology to solve real problems strategically for real organization.
CLO6. Demonstrate 21st century skills in all learning activities.

LEARNING PLAN:

Learning Outcomes for Midterm Period

Course Learning Outcomes	Topic Learning Outcomes
CLO 1. Understand the latest concepts in systems analysis and design, software engineering and information technology.	TLO1. Analyze complex problems, identify, and define the computing requirements appropriate to its solution. TLO2. Identify and analyze user needs and take them into account in the selection, creation, and evaluation and administration of computer-based systems. TLO3. Study and apply various high level and detailed design and implementation principles for building a software system.
CLO 2. Analyze original research data and present the data in a form appropriate for the design.	TLO4. Design, implement, and evaluate computer-based systems, processes, components, or programs to meet desired needs and requirements under various constraints.
CLO 3. Collect original data following ethical protocols.	TLO5. Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information technology.
CLO 4. Interpret the data collected, draw conclusions solidly rooted in the data, and convey those interpretations and implications in a formal written report.	TLO6. Function effectively as a member or leader of a development team recognizing the different roles of a team to accomplish a common goal. TLO7. Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and clear instructions.
CLO 5. Design and implement integrated information systems and technology to solve real problems strategically for real organization.	TLO8. Apply knowledge through the use of current techniques, skills, tools, and practices necessary for the IT profession. TLO9. Develop a formal Presentation of the research project and engage/answer questions in an oral presentation
CLO 6. Demonstrate 21st century skills in all learning activities.	TLO10. Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)

Module Number and Learning Outcome/s	Date	Teaching-Learning Lessons and Activities / Assessment Strategies Online Learning Class Lessons and Activities	Technology Enabler	Onsite	Asynchronous	Allotted Hours
Module 1 CLO1 CLO6 TLO1 TLO2 TLO3 TLO10	Week 1 Aug 19-24	**Course Orientation** • Gospel Reading, Reflection and Prayer • Course Overview • Course Polices and Requirements • Professor Profile • Grading System • Netiquette • Group meeting	Schoolbook MS Teams Power Point	1.5	1.5	3.0
Module 2 CLO2 CLO3 CLO6 TLO4 TLO5 TLO10	Week 2 Aug 26-31	**Chapter 4 Results and Findings** • Gospel Reading, Reflection and Prayer • Writing the results and findings • Cross Validation • Data Validation and Consultation with the Adviser	Schoolbook MS Teams Power Point	1.5	1.5	3.0
		Formative Assessment: Submission of Progress Report	Schoolbook			
Research Work CLO2 CLO3 CLO4 CLO5 CLO6 TLO4 TLO5 TLO6 TLO7 TLO8 TLO9 TLO10	Week 3-5 Sep 2 - 21	• Gospel Reading, Reflection and Prayer • Journal Manuscript Writing • System Development • Consultation with the Adviser	Schoolbook MS Teams		9.0	9.0

Module 3 CLO3 CLO4 TLO5 TLO6 TLO7	Week 6 Sep 23 - 28	**Chapter 5 Summary, Conclusions and Recommendations** • Gospel Reading, Reflection and Prayer • Writing the summary, conclusions and recommendations **Formative Assessment:** Submission of Progress Report and Ethics Certificate	Schoolbook MS Teams Power Point Schoolbook	1.5	1.5	3.0
Research Work CLO2 CLO3 CLO4 CLO5 CLO6 TLO4 TLO5 TLO6 TLO7 TLO8 TLO9 TLO10	Week 7-9 Sep 30 – Oct 19	• Gospel Reading, Reflection and Prayer • Journal Manuscript Writing • System Development • Consultation with the Adviser **Formative Assessment:** Submission of Progress Report	Schoolbook MS Teams		9.0	9.0
			Total	**4.5**	**22.5**	**27**

Learning Outcomes for Final Period

Course Learning Outcomes	Topic Learning Outcomes
CLO3. Collect original data following ethical protocols.	TLO5. Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information technology.
CLO4. Interpret the data collected, draw conclusions solidly rooted in the data, and convey those interpretations and implications in a formal written report.	TLO6. Function effectively as a member or leader of a development team recognizing the different roles of a team to accomplish a common goal. TLO7 Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and clear instructions.
CLO5. Design and implement integrated information systems and technology to solve real problems strategically for real organization.	TLO8. Apply knowledge through the use of current techniques, skills, tools, and practices necessary for the IT profession. TLO9. Develop a formal Presentation of the research project and engage/answer questions in an oral presentation.

CLO6. Demonstrate 21st century skills in all learning activities.	TLO10. Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)

Module Number and Learning Outcome/s	**Date**	**Teaching-Learning Lessons and Activities / Assessment Strategies** **Online Learning Class Lessons and Activities**	**Technology Enabler**	**Onsite**	**Asynchronous**	**Allotted Hours**
Module 4 CLO4 TLO6 TLO7	Week 10 Oct 21-26	**Publishing Paper – Article Writing** • Gospel Reading, Reflection and Prayer • IMRAD Guide • Article Writing	Schoolbook MS Teams	1.5	1.5	3.0
	Week 11 Oct 28 – Nov 2	Independent Learning Week			3.0	3.0
Research Work CLO3 CLO4 CLO5 CLO6 TLO5 TLO6 TLO7 TLO8 TLO9 TLO10	Week 12–14 Nov 4 - Nov 23	• Gospel Reading, Reflection and Prayer • Consultation with Advisers • Journal Manuscript Writing • System Development **Enabling assessment:** Submission of Progress Report and Endorsement Form of Adviser	Schoolbook		9.0	9.0
Final Defense CLO5 CLO6 TLO8 TLO9 TLO10	Week 15–18 Nov 25 - Dec 18	**Summative Assessment**: Oral Presentation (Final)		12.0		12.0
			Total	**13.5**	**13.5**	**27**

GRADING SYSTEM:

Finals
Enabling Assessments (Adviser's Grade) - 20%
Summative Assessments (Panel's Grade) - 80%
100%

COURSE POLICIES AND REQUIREMENTS:

1. **Office365 Activation.** Please ensure that your Office365 account is working. Your Office365 account is needed to access both Schoolbook and MS Teams where your asynchronous and synchronous classes will be held.

2. **Enrollment in an E-Class.** You will automatically be enrolled in your e-class which is based on your enrollment data.

3. **Traditional Blended Modality.** This course adopts the traditional blended model. This means that there will be a mix of face-to-face, and asynchronous activities or online classes. This is divided into fifty percent face-to-face and fifty percent online.

4. **Online Asynchronous Sessions**.
    a. **Schoolbook (SB).** Schoolbook shall be the only platform for asynchronous sessions.
    b. Modules. Modules are self-paced learning resources for asynchronous sessions. These can be accessed in Schoolbook.
    c. **References.** Each page section may contain uploaded references. These learning resources may be downloaded.
    d. **Asynchronous Activities.** You are expected to read the modules as soon as they are uploaded. The learning content of the modules complement the online synchronous and face-to-face sessions.
    e. **Asynchronous Engagement.** Your activities in the course can be tracked by your professor. This includes the time you spend in reading the lessons and answering the assessments.
    f. **Schoolbook Forum.** All general concerns about the lessons and assessments in asynchronous sessions must be posted in the Schoolbook Forum. Response shall be made by your teacher within 48 hours.
    g. **Schoolbook Messaging.** This shall be the mode of communication for private and/or confidential communications. Response shall be made by your teacher within 48 hours upon receipt of the same unless it falls on weekends or holidays, which shall be handled promptly the following working day.

5. **Onsite / Face-to-face (F2F) Sessions.**
    a. **Face-to-face engagement.** Your engagement in face-to-face classes is graded based on your class participation.

b. **Classroom.** F2F classes shall be held at the classroom indicated in your Certificate of Registration. Should there be changes in the classroom venue, information will be given in advance.
c. **Gospel Reading and Prayer.** Each F2F session shall start with a Gospel reading and prayer. Your teacher may assign you, in advance, to do this.
d. **F2F Meeting Schedule.** The meeting schedule shall follow the time indicated in your official registration. The dates of F2F meetings are identified in the learning plan.
e. **Attendance.** Attendance in F2F meetings is required.
f. **Tardiness.** A student who came 1 minute to 45 minutes after the start of the face-to-face meeting is considered late. Three tardy attendance is equivalent to 1 absence.
g. **Absence.** A student is considered absent 45 minutes after the official class schedule.
h. **Excuse from F2F classes.** Students are excused in the F2F classes based on the provisions in the latest version of the Student Handbook.
i. **Dress code.** Attire should follow the policy on dress code as stipulated in the latest version of the Student Handbook.

6. **Assessment and Grading System.**
   a. **Formative assessments.** These are ungraded assessments. These may be considered as practice assessments that leads towards achieving outcomes without fear of receiving a failing grade.
   b. **Enabling assessments.** These will comprise most of your graded assessments. These are designed to achieve topic learning outcomes, that leads towards achieving the course learning outcomes. Only one enabling assessment shall be allowed during the week. Please pay attention to the duration and number of attempts. As a general rule, quiz-type enabling assessments shall be open for only a minimum of 24 hours, while output-based enabling assessments shall be open for at least 6 days.
   c. **No. of Attempts.** All online enabling assessments shall have a maximum of 3 attempts only. Onsite enabling/summative assessments shall have 1 attempt only.
   d. **Summative assessments.**

      i. There shall be two summative assessments (midterm and final exams) for the entire semester. These are designed to achieve the course learning outcomes.
      ii. Summative assessment is administered face-to-face.
      iii. Output-based summative assessment shall be given to students at least fifteen days prior to scheduled Summative Exam Week.

   e. **Lifeline.** Only students with (1) valid reason as stated in the Student Handbook and IRR, and (2) given their proof of excuse on or before the next synchronous/F2F session, shall be given a lifeline on the enabling and summative assessments.
   f. **Rubric.** All online non-quiz or non-discrete types of assessment (essay, drop box, output-based, etc.) shall have a rubric or criteria for rating the students' tasks. A student may refuse to answer these types of assessment in the absence of a rubric or criteria for grading, and the assessment shall be deemed invalid and shall not be part of the student's grades.

g. **Grading.** All online assessments should be checked and graded by the teacher before the submission of midterm and final grades.
h. **Gradebook.** Students can see the breakdown of grades in their Assessment tab.

7. **Self-Care.**
    a. **Schedule.** Self-care Week is set after the midterm examination week. There shall be no synchronous meetings, F2F classes, new modules, new assessments, and deadlines on this date.
    b. **Prerogative.** Students may avail of the self-care program, whether online or onsite, provided by the different units of the University.

8. **Data Privacy.**
    a. **On camera.** Your teacher may require you to engage in synchronous session by turning on your camera. These type of engagement activities include, but not limited to the following: chat participation, participating by audio/voice, participation in poll and games, clicking links, and other online engagement activities.
    b. **Recording of synchronous session.** Your teacher shall inform you, at the start of the meeting, that the synchronous meeting is being recorded. If you do not consent on the recording, please provide a valid reason.
    c. **Access to the MS Teams recording.** Only students who are officially enrolled shall be part of the MS Teams and have access to all the resources including the recording. Students are not allowed to record the meetings. Students are not allowed to download the recordings. Screen recording is not allowed.
    d. **Guests.** Inviting people, that are not part of the class, in synchronous meetings is strictly prohibited, unless approved by the subject teacher.
    e. Posing as another person during a synchronous activity is strictly prohibited.

9. **Copyright and Plagiarism.**
    a. Using of any illegally obtained software and other technology tool is strictly prohibited.
    b. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
    c. Giving of password to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 account other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).
    d. This subject shall abide by the policies pertaining to intellectual property, copyright, and plagiarism as stipulated in latest edition of the Student Handbook.
    e. Any plagiarized work, whether in part or full, shall mean a grade of 0.0 for the assessment.

10. This course shall abide by any institutional policies that may be released after the approval of this syllabus. Any such policy shall be posted within the e-class at the forums section, news feed. It will also be briefly discussed during the soonest synchronous meeting.

REFERENCES:

Call number or e-provider	References
	Undergraduate Research Handbook: Policies and Guidelines
QA 402 .D423a 2019	Dennis A, Wixom B, & Roth R. 2019. Systems analysis and design. 7th edition. Hoboken, NJ : Wiley
QA 76.9.D26 .C816 2019	Coronel C, Morris S. 2019. Database systems: design, implementation and management. 13th edition. (Australia, United States): Cengage Learning
Journals	Z. Gao, M. P. Gatpandan and P. H. Gatpandan, "Classification Decision Tree Algorithm in Predicting Students' Course Preference," *2021 2nd International Symposium on Computer Engineering and Intelligent Communications (ISCEIC)*, Nanjing, China, 2021, pp. 93-97, doi: 10.1109/ISCEIC53685.2021.00026. J. Xu, P. H. Gatpandan, M. P. Gatpandan and Z. Gao, "Freight and Passenger Volume Prediction using Multiple Regression," *2021 International Conference on Machine Learning and Intelligent Systems Engineering (MLISE)*, Chongqing, China, 2021, pp. 77-83, doi: 10.1109/MLISE54096.2021.00022. X. Gao, M. F. Rosas, Y. Zhang, M. P. Gatpandan, P. H. Gatpandan and D. Dasig, "The Research and Analysis of Student Health Status on the Basis of Physic Fitness Test' Data," *2021 International Conference on Public Health and Data Science (ICPHDS)*, Chengdu, China, 2021, pp. 110-115, doi: 10.1109/ICPHDS53608.2021.00031. D. J. R. Calantoc, D. D. Dasig, E. E. Claricia, M. A. B. Taduyo, P. H. Gatpandan and R. V. F. Guarin, "Clustering and Predicting Smartphones Features using Gaussian Mixture Model Algorithm," *2021 IEEE 13th International Conference on Humanoid, Nanotechnology, Information Technology, Communication and Control, Environment, and Management (HNICEM)*, Manila, Philippines, 2021, pp. 1-6, doi: 10.1109/HNICEM54116.2021.9731964.

Prepared: **Information Technology Department**
1st Semester 2024-2025

Approved: **(SGD) Ms. Sherry B. Naz**
Chair, Information Technology Department

Endorsed: **(SGD) Ms. Marivic R. Mitschek**
Dean, College of Information and Computer Studies

cf: OVCAR, AEA, CICS Curriculum Development Committee, file